# Jui Roy

## Data Analyst

I turn messy data into decisions people act on. Four case studies built with real public datasets, using SQL, Python and Excel.

1. **Superstore sales and profit**
   Python / SQL, 9,994 order lines

2. **Telecom customer churn**
   Python / SQL, 7,043 customers

3. **HR attrition dashboard**
   Excel, 1,470 employees

4. **Netflix catalogue analysis**
   Excel, 6,234 titles

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CASE STUDY 1 / PYTHON AND SQL

# Superstore: where does the business lose money?

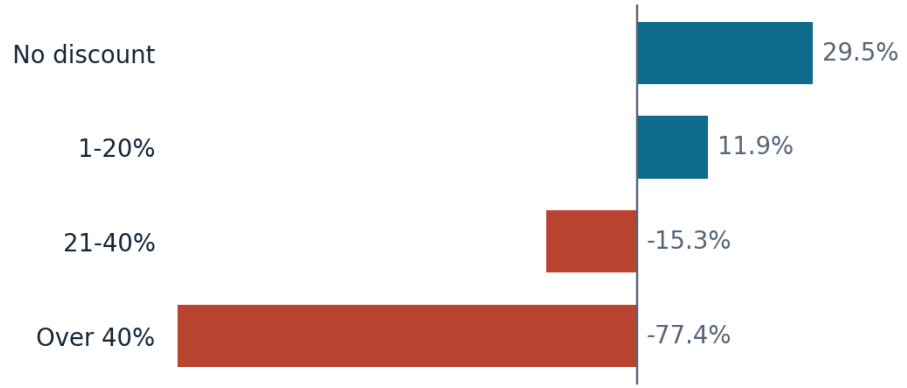
Sales grew every year after 2015. Is the business actually profitable, and which products and discount levels leak profit?

**$2.30M**
Sales, 2014-2017

**$286.4K**
Profit

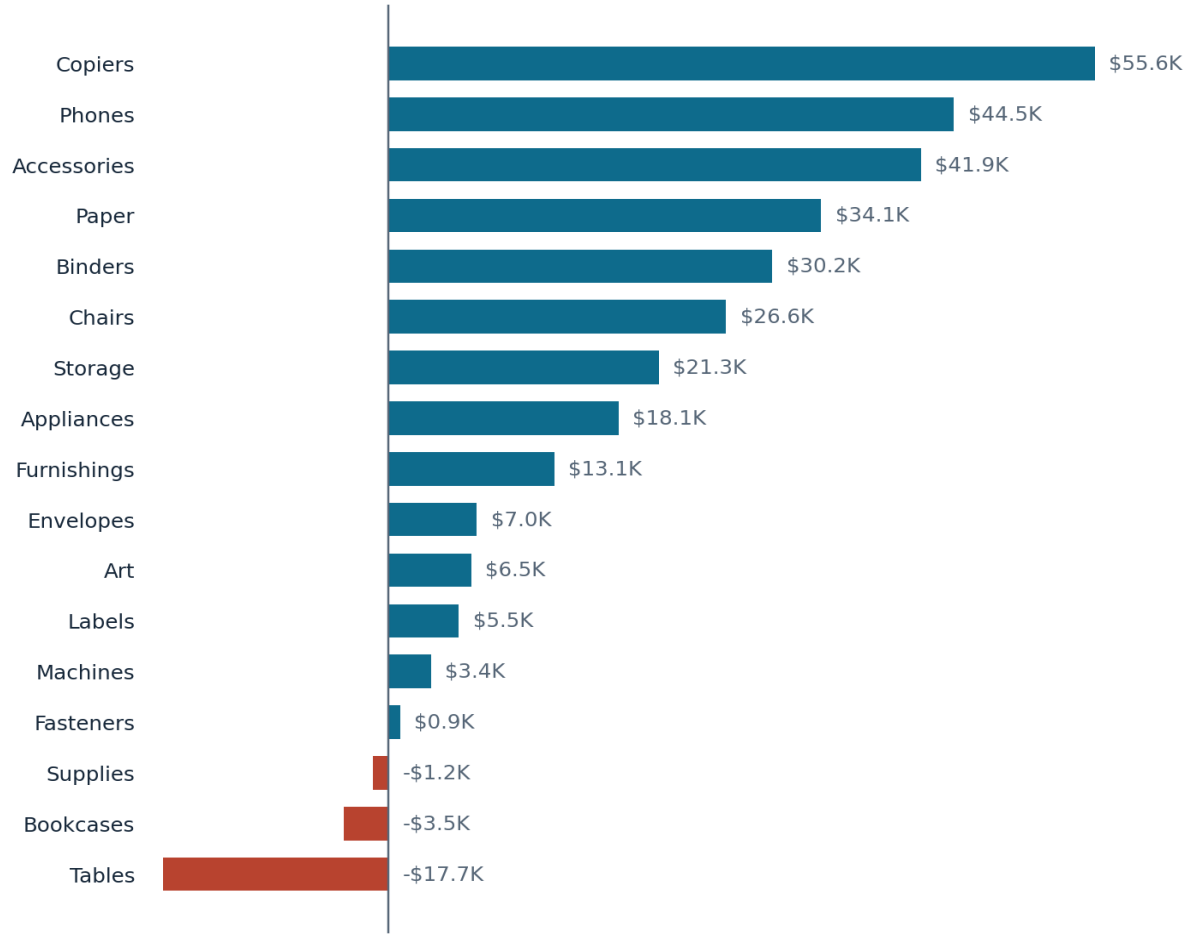
**12.5%**
Profit margin

**Profit margin by discount level**




- Order lines discounted over 40% (933 of 9,994) run at a margin of about -77%; undiscounted lines earn 29.5%.
- Tables, Bookcases and Supplies lose money despite real revenue (Tables: about -$17.7K).
- Recommendation: cap discounts near 20%, test it for one quarter, and measure margin and volume.

**Profit by product sub-category**



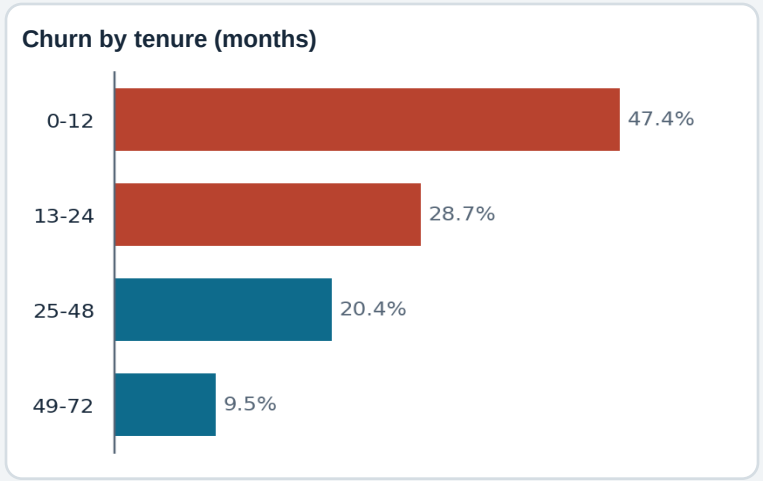
CASE STUDY 2 / PYTHON AND SQL

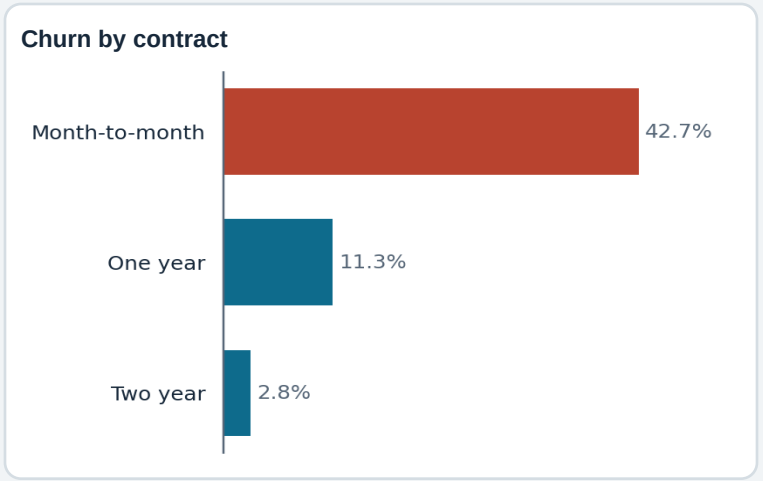
# Telecom churn: who leaves, and what does it cost?

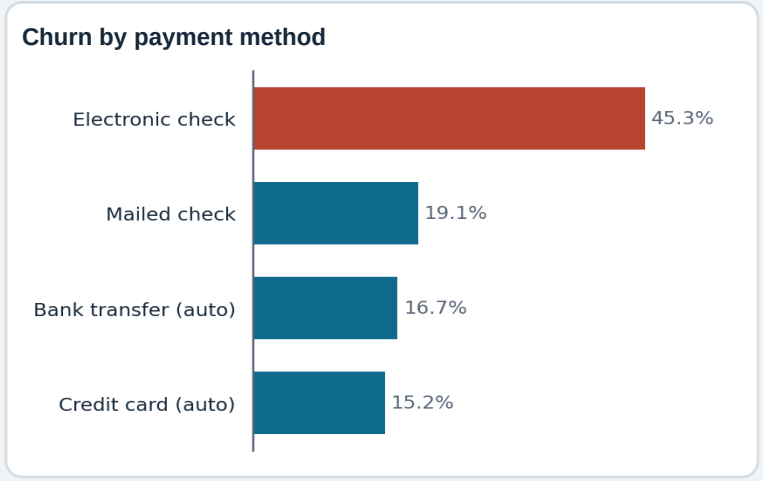
About one in four customers cancels. Red bars are above the 26.5% average.

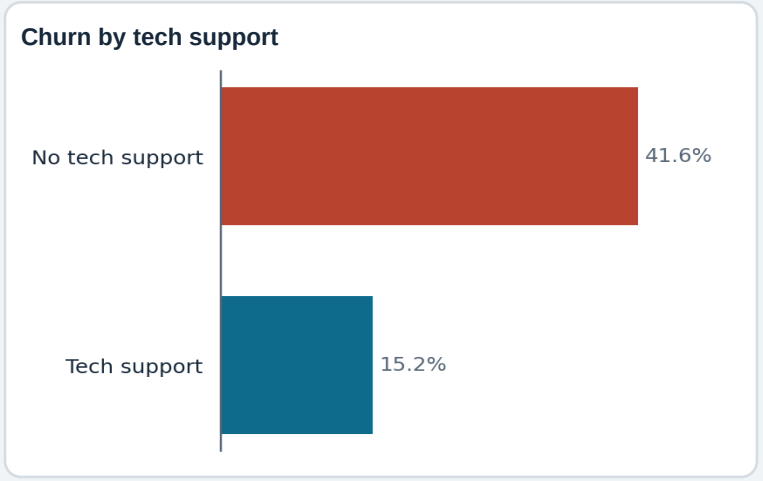
| 7,043 | 26.5% | $139.1K |
|---|---|---|
| Customers | Churn rate | Monthly revenue lost |

- Month-to-month customers churn at 42.7%, versus 2.8% on two-year contracts.
- Customers in their first 12 months churn at 47.4%; after four years, 9.5%.
- Electronic check users churn at 45.3%, versus 15-17% on automatic payments.
- Customers without tech support churn at 41.6%, versus 15.2% with it.
- Recommendation: offer first-year customers an annual contract with autopay and bundled tech support.



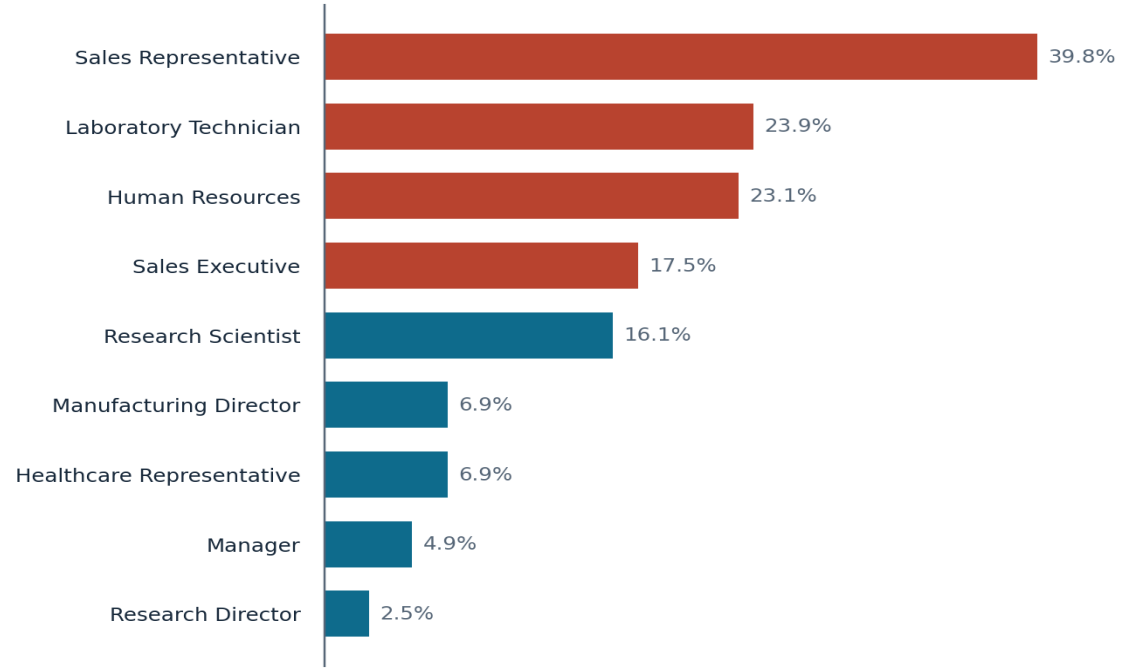
PROJECT 3 / BUILT ENTIRELY IN EXCEL

# HR attrition dashboard

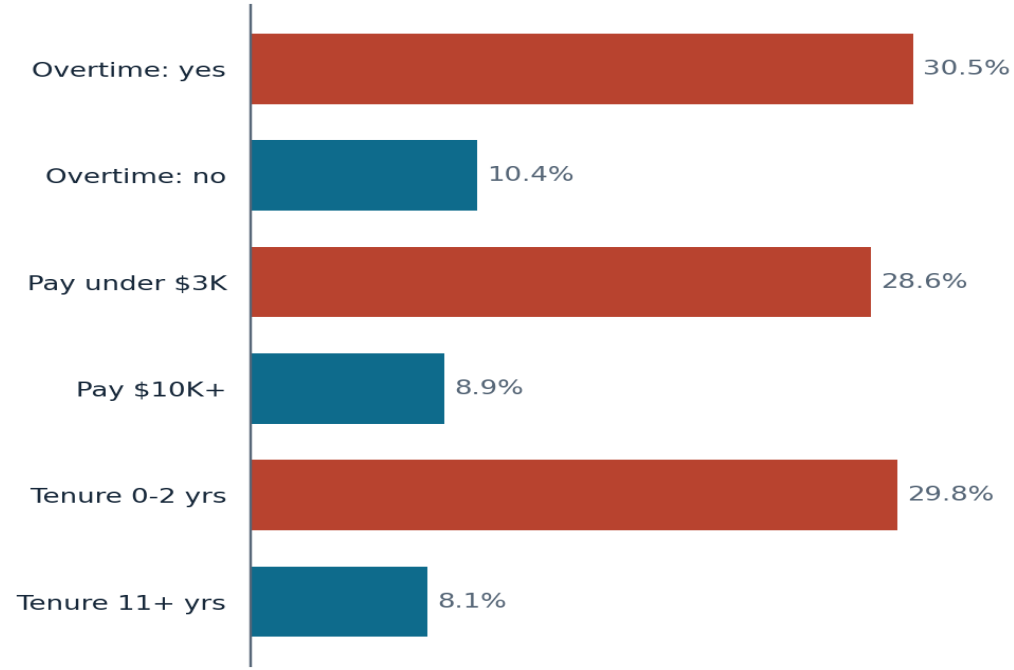
IBM HR sample data. Live formulas (COUNTIFS, SUMIFS, IFERROR), banded fields, 4 native charts and findings written with formulas.

| 1,470 | 16.1% | 237 |
|---|---|---|
| Employees | Attrition rate | Leavers |

**Attrition rate by job role**




**Attrition by group (red = above the average)**




Overtime workers leave at 30.5% versus 10.4%. Sales Representatives leave at 39.8%. Staff on under $3K a month leave at 28.6%, and the first two years are the riskiest.

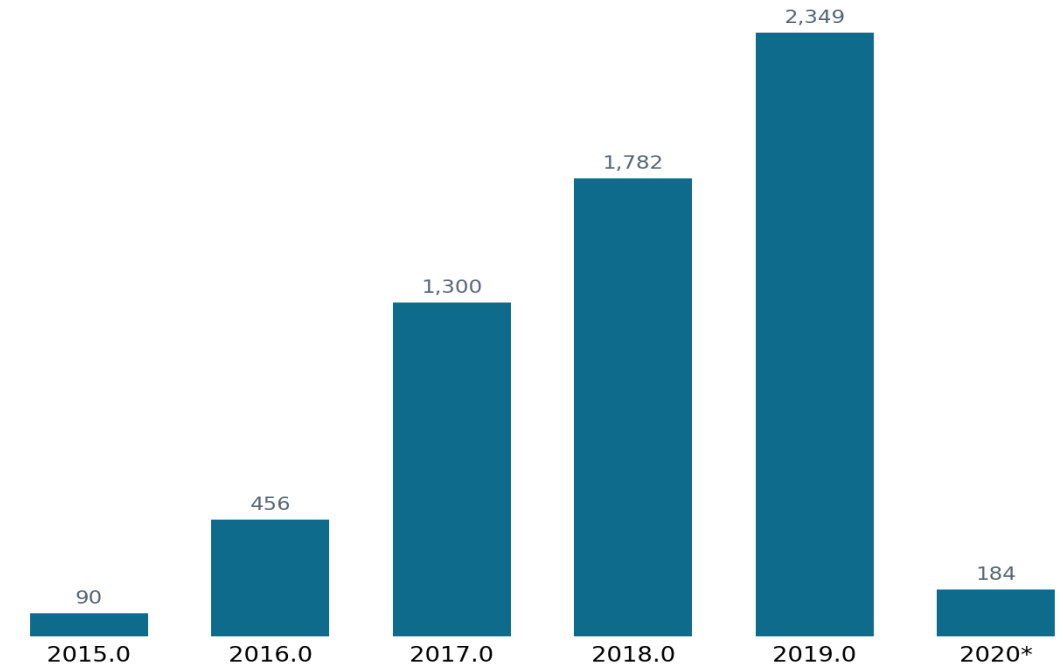
PROJECT 4 / BUILT ENTIRELY IN EXCEL

# Netflix catalogue: cleaning and analysis

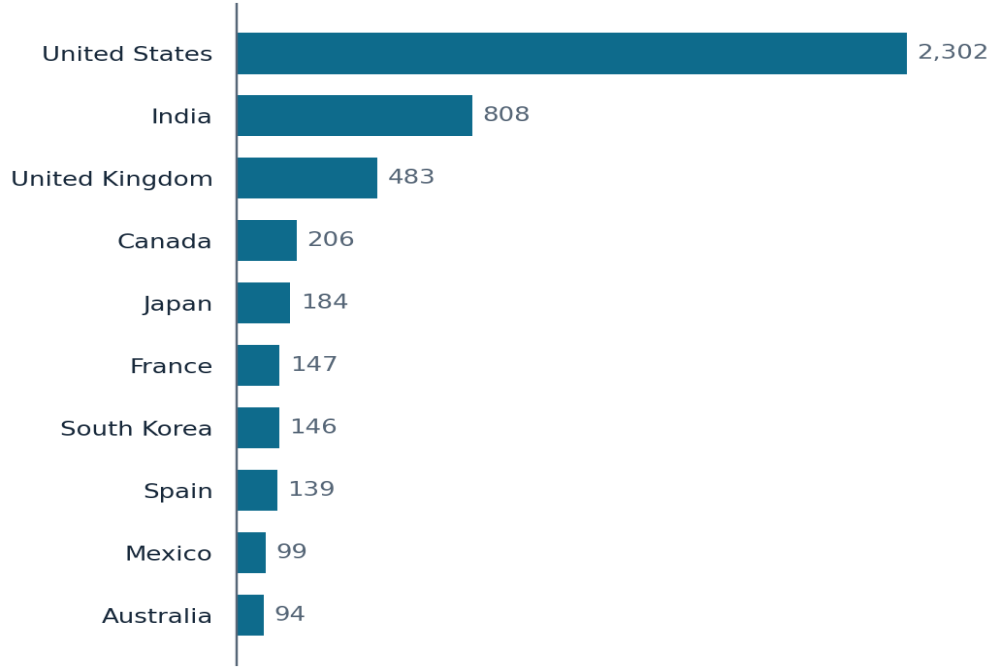
Public Netflix titles file cleaned with formulas only (TRIM, LEFT, RIGHT, FIND, VALUE, COUNTBLANK), about 37,000 formulas, a cleaning log and 4 charts.

| 6,234 | 68% | 99 min |
|---|---|---|
| Titles | Movies | Average movie |

**Titles added per year (*2020 is a partial year)**




**Top 10 countries (titles)**




Additions rose from 90 titles in 2015 to 2,349 in 2019. The US accounts for 36.9% of titles and Dramas is the top genre. 31.6% of titles have no director listed.

SKILLS AND CONTACT

# Tools I use, and how to reach me

## Querying and cleaning

- SQL: joins, window functions, CTEs
- Python: pandas, NumPy
- Advanced Excel: COUNTIFS, SUMIFS, INDEX/MATCH

## Analysis

- Segmentation and cohort thinking
- Churn and profitability analysis
- Statistics and forecasting basics

## Communicating

- Excel dashboards and charts
- Matplotlib and Plotly visuals
- Plain-language findings and recommendations

## Open to data analyst roles and freelance projects


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